

Challenge (Habanero)

- Q1.** A hiker travels 3.5 miles in 2.5 hours. What is the hiker's unit rate in miles per hour?
- (A) 1.4 miles/hour
(B) 1.7 miles/hour
(C) 1.2 miles/hour
(D) 1.0 mile/hour
(E) 0.8 miles/hour
- Q2.** A car's fuel consumption is 25 miles per gallon. How many gallons will it consume to travel 375 miles?
- (A) 12 gallons
(B) 15 gallons
(C) 10 gallons
(D) 8 gallons
(E) 20 gallons
- Q3.** The population of rabbits in a forest is directly proportional to the amount of food available. If 50 rabbits require 200 pounds of food, how many rabbits can be supported with 480 pounds of food?
- (A) 80 rabbits
(B) 100 rabbits
(C) 120 rabbits
(D) 150 rabbits
(E) 240 rabbits
- Q4.** A group of friends want to divide some trail mix equally. If they have 18 cups of trail mix and there are 6 friends, how many cups will each friend get?
- (A) 2 cups
(B) 3 cups
(C) 1 cup
(D) 4 cups
(E) 6 cups
- Q5.** A bicycle travels at a rate of 15 miles per hour. How many hours will it take to travel 45 miles?
- (A) 1 hour
- (B) 2 hours
(C) 3 hours
(D) 4 hours
(E) 5 hours
- Q6.** The cost of a hike is directly proportional to the number of hikers. If a hike for 8 people costs 160, how much will a hike for 12 people cost?
- (A) 240
(B) 260
(C) 280
(D) 300
(E) 320
- Q7.** A car travels 250 miles in 5 hours. What is the unit rate in miles per hour?
- (A) 40 miles/hour
(B) 50 miles/hour
(C) 48 miles/hour
(D) 55 miles/hour
(E) 60 miles/hour
- Q8.** The time it takes to complete a hike is inversely proportional to the number of hikers. If a hike for 4 people takes 6 hours, how many hours will a hike for 8 people take?
- (A) 2 hours
(B) 3 hours
(C) 1.5 hours
(D) 4.5 hours
(E) 6 hours

Open Ended Questions (FRQ)

- Q9.** A park ranger is planning a hiking trip. The trip will cover a total distance of 12 miles and the ranger wants to maintain a constant speed of 2.5 miles per hour. How many hours will the trip take? Show your work and calculate the time it will take to complete the trip. Assume the hikers do not take any breaks.
- Q10.** The population of deer in a forest is directly proportional to the amount of food available. If 80 deer require 400 pounds of food, how many pounds of food will be needed to support 120 deer? Use the concept of direct variation to solve the problem and show your work.

Chef's Tips

- Emphasize the importance of reading the questions carefully
- Allow students to use calculators for calculations
- Remind students to show their work for free response questions